HAMMERSPACE

MLPerf Storage

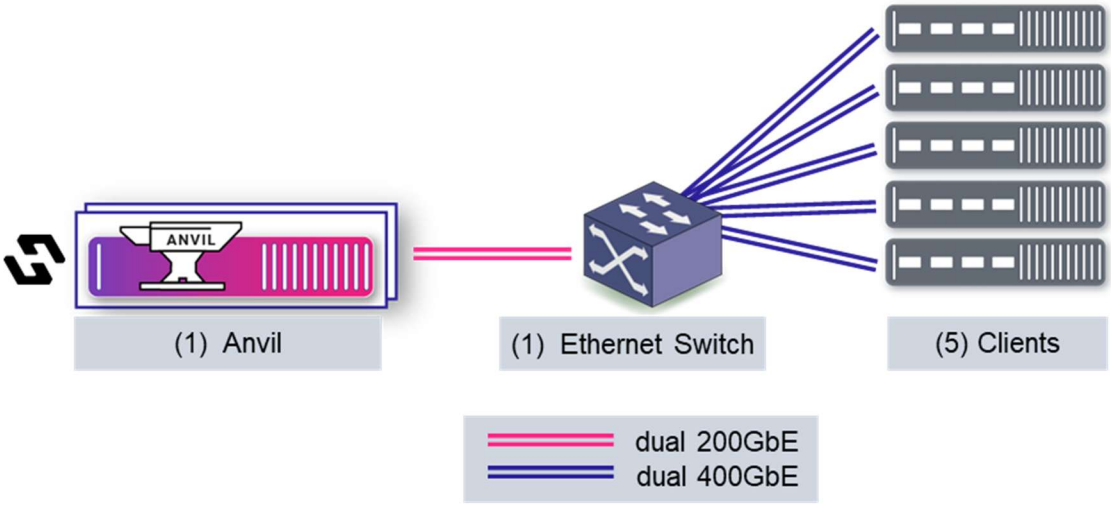
System Description

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System Name:	Hammerspace_5Tier0_H100
Submitter Name:	Hammerspace
Submission Date:	July 11, 2025
Benchmark Version:	2.0
Solution Type:	Shared File with Client Local NVMe
Submission Division:	CLOSED
Training Workloads:	3D U-Net H100

This document describes the Hammerspace test environment used to generate a subset of the submitted results for the MLPerf Storage v2.0 Benchmark Suite, per the submission guidelines.

System Topology



For these test runs the Hammerspace system consists of one Anvil metadata node, one Ethernet switch, and five clients, connected as shown in the figure above. The Anvil is responsible for metadata operations and cluster coordination tasks. The clients run the benchmark code and also act as storage servers. Internal NVMe drives in each client become elements of the Hammerspace shared file system coordinated by the Anvil. This is an example of the configuration referred to in Hammerspace marketing as “Tier 0.”

Clients mount the shared file system via parallel NFS (pNFS) v4.2 and use the Flexible Files layout type. The specific mount command used is listed in the Settings section below. To perform I/O, clients first contact the pNFS metadata server (the Anvil) to obtain a *layout*. Once in possession of a layout, clients directly communicate with the storage servers. Data does not flow through the metadata server. This is standard pNFS behavior.

Hardware

The hardware configuration used for these runs is detailed in the tables below.

Anvil

Resource	Qty	Specification
System	1	Lenovo SR630 v3
CPU	2	Intel Xeon Gold 6542Y 60MB Cache, 2.90 GHz, 48-Core
Memory	16	Samsung M321R4GA3PB0-CWMXJ 32GB DDR5-5600 ECC RDIMM Total 512GB RAM
Boot Drive	2	Samsung MZQL21T9HCJR-00A07 1.92TB PCIe4 x4 U.2 SSD (Mirrored)
Network Adapters	1	NVIDIA CX755106AS-HEAT ConnectX-7 Dual 200GbE QSFP112 PCIe5 x16
Storage Drives	2	ScaleFlux CSD5000 7.68TB NVMe PCIe5 x4 U.2 (Metadata - Mirrored)

Table 1 - Anvil (metadata server node) Hardware Specification

MLPerf Client

Resource	Qty	Specification
System	1	Supermicro SYS-121C-TN10R
CPU	2	Intel Xeon Gold 6542Y 60MB Cache, 2.90 GHz, 48-Core
Memory	16	Micron MEM-DR564L-CL02-ER56 64GB DDR5-5600 ECC RDIMM, Total 1TB RAM
Boot Drive	2	Micron 7450 Pro MTFDKBG1T9TFR 1.92 TB NVMe PCIe4 M.2 TLC
Network Adapters	2	NVIDIA MCX75310AAS-NEAT ConnectX-7 Single 400GbE OSFP PCIe5 x16
Storage Drives	10	ScaleFlux CSD5000 7.68TB NVMe PCIe5 x4 U.2 (RAID-0)

Table 2 - MLPerf Client Hardware Specification

Network

Resource	Qty	Specification
Ethernet Switch	1	Supermicro SSE-T8032S Twin-port 2x 400GbE Open Networking Switch

Table 3 - Network Hardware Specification

Storage Rack Units

Resource	Qty	Rack U each	Total Rack Units
Anvil	1	1	1
Configuration Total			1

Table 4 - Rack Unit Summary

Power Requirements

System Component	PS Unit	Nameplate Power	Design Power
Anvil	1	1100W	1100W
	2	1100W	
MLPerf Client #1	1	860W	860W
	2	860W	
MLPerf Client #2	1	860W	860W
	2	860W	
MLPerf Client #3	1	860W	860W
	2	860W	
MLPerf Client #4	1	860W	860W
	2	860W	
MLPerf Client #5	1	860W	860W
	2	860W	
		Configuration Total	5400W

Table 5 - Power Requirements

Software

The software running on each component type in this configuration is detailed below.

Anvil

Hammerspace is packaged as a single installation ISO that includes a Linux operating system, application code, and all dependencies.

- Hammerspace: v5.2.2-39
- Kernel: 6.12.24-8.hs.91.el8.x86_64
- Pd-protod RPM: pd-protod-5.2.2-86124.el8.x86_64.rpm

MLPerf Client

- Standard, unmodified MLPerf benchmark v2.0 code
- Rocky Linux v9.5 kernel 6.12.24-8.hs.91.e19.x86_64

Network Switch

- SONiC OS v4.5.0 Enterprise Advanced.

Settings

Anvil

- Increased the number of initial instances created using this setting in the /pd/dme/configuration.ini file:

  [layoutManager]
  maxInitialNASInstances = 5

MLPerf Client

The following changes were made to settings for the NFS client:

- /sys/module/nfs/parameters/localio_async_probe (Default Y) Setting is N
- /sys/module/nfs/parameters/localio_enabled (Default is N) Setting is Y
- /sys/module/nfs/parameters/localio_O_DIRECT_semantics (Default N) Setting is Y
- /sys/module/nfsv4/parameters/nfs4_files_uncacheable (Default N) Setting is Y

The following changes were made to settings for the NFS server:

- /sys/kernel/debug/nfsd/io_cache_read (Default 0) Setting is 2
- /sys/kernel/debug/nfsd/io_cache_write (Default 0) Setting is 2

Clients mounted the shared file system using this command:

```
mount -t nfs -o nconnect=16,port=20491 192.168.0.161:/hs_test /mnt/hs_test
```

Network Switch

No tunings were applied to the network switch.

Miscellaneous

Anvil

Hammerspace enables “Objectives” to be set by administrators. Objectives are declarative policies describing desired data-handling behavior, including aspects such as data durability.

For this configuration one objective was set, specifying data durability of nine nines.